

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Sundar Raj	Report ID:	APL-IRON-19-SUN-2026
Age / Gender:	19 Yrs / Male	Category:	Iron Deficiency Anaemia
Weight / Height:	58 kg / 170 cm	BMI:	20.1 kg/m2
Ref. Doctor:	Dr. Arjun Mehta	Report Date:	12-Jan-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	8.2	g/dL	13.0 - 17.5	LOW
Packed Cell Volume (PCV)	31.6	%	40 - 52	LOW
MCV	75.8	fL	83 - 101	LOW
MCH	23.7	pg	27 - 32	LOW
RDW-CV	16.7	%	11.6 - 14.0	HIGH

Haematology Comment: Haemoglobin 8.2 g/dL - microcytic (MCV 75.8 fL) hypochromic (MCH 23.7 pg) anaemia pattern. Raised RDW-CV (16.7%) indicates anisocytosis. Consistent with iron deficiency or thalassaemia trait. Further iron studies recommended.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	7000	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	48	ug/dL	60 - 170	LOW
TIBC	390	ug/dL	250 - 370	HIGH
Transferrin Saturation	13.9	%	20 - 50	LOW
Serum Ferritin	6.0	ng/mL	30 - 300	LOW

Iron Studies - Critical Findings: Serum iron 48 ug/dL and ferritin 6 ng/mL are reduced. Transferrin saturation 13.9% (normal >20%).
RECOMMENDATION: Iron supplementation 60 mg elemental iron/day. Investigate dietary adequacy and occult GI blood loss.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	22	U/L	7 - 56	Normal
SGOT / AST	20	U/L	10 - 40	Normal
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	4.2	g/dL	3.5 - 5.0	Normal

Hepatologist Note: Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	0.9	mg/dL	0.7 - 1.3	Normal
Blood Urea Nitrogen	14	mg/dL	7 - 20	Normal
eGFR (CKD-EPI)	98	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 7.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 98 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	172	mg/dL	0 - 200	Normal
HDL Cholesterol	55	mg/dL	40 - 999	Normal
LDL Cholesterol	98	mg/dL	0 - 100	Normal
Triglycerides	88	mg/dL	0 - 150	Normal

Cardiologist Note: Lipid profile within optimal range. Cardiovascular risk from dyslipidaemia is low.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	2.1	uIU/mL	0.4 - 4.0	Normal
Free T3	3.2	pg/mL	2.3 - 4.2	Normal
Free T4	1.1	ng/dL	0.8 - 1.8	Normal

Endocrinologist Comment: Thyroid function normal. TSH 2.1 uIU/mL. Euthyroid state confirmed.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	88	mg/dL	70 - 100	Normal
Post-Prandial Glucose (2hr)	115	mg/dL	0 - 140	Normal
HbA1c	5.1	%	0 - 5.7	Normal
HOMA-IR	1.5		0 - 2.5	Normal

Endocrinology Note: Glucose metabolism normal. HbA1c 5.1% - no diabetes or pre-diabetes. HOMA-IR 1.5 - no insulin resistance.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	20.7	ng/mL	30 - 100	LOW
Vitamin B12	450	pg/mL	211 - 911	Normal
Folate (Serum)	12.0	ng/mL	5.4 - 999	Normal
Calcium (Total)	9.4	mg/dL	8.5 - 10.5	Normal
PTH	32	pg/mL	15 - 65	Normal

Nutritionist/Endocrinology Note: Vitamin D 21 ng/mL - insufficient (20-30). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	31	mm/hr	0 - 15	HIGH
CRP	3.9	mg/L	0 - 5	Normal
hs-CRP	1.4	mg/L	0 - 1.0	HIGH

Immunology Comment: CRP 3.9 mg/L and ESR 31 mm/hr are elevated. hs-CRP 1.4 mg/L - intermediate CV risk (1-3). Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

- [ABNORMAL]** Anaemia: Hb=8.2 g/dL (LOW). Iron deficiency pattern.
- [CRITICAL]** Iron Deficiency: Serum Iron=48 ug/dL, Ferritin=6 ng/mL. Begin iron supplementation.
- [DEFICIENT]** Vitamin D=21 ng/mL (LOW). Supplementation required.
- [ACTION]** Next appointment: Dr. Arjun Mehta in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Arjun Mehta

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